A Review of Smooth Estimation of the Cumulative Distribution Function under Judgment Post Stratification

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Abstract:

This paper provides a review of smooth estimation methods for the cumulative distribution function (CDF) under the judgment post stratification (JPS) sampling scheme. Particular attention is devoted to a general class of CDF estimators that encompasses both empirical and kernel-based estimators. The main finite-sample and asymptotic properties of these estimators are summarized and discussed. To further illustrate their performance, an extended Monte Carlo simulation study is conducted to compare the kernel distribution function (KDF) estimator under JPS with its counterpart based on simple random sampling (SRS). The comparison considers a range of sample sizes, set sizes, and ranking qualities. The findings demonstrate that incorporating ranking information through the JPS design can lead to considerable efficiency gains relative to SRS across a broad spectrum of settings.

Keywords: Nonparametric estimation, Monte Carlo simulation, Judgment ranking, and Statistical inference

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1 Introduction

Judgment post stratification (JPS) sampling, proposed by MacEachern et al. (2004), is a rank-based sampling design that utilizes ranking information to improve the efficiency of statistical inference. This sampling scheme is particularly useful in studies where accurate measurement of the variable of interest (X) is difficult, expensive, or time-consuming, while ranking the sampling units can be performed easily and at a relatively low cost. Such situations frequently arise in ecological, medical, and environmental research.

To construct a JPS sample of size n, one first determines the set size H. Then, one draws a simple random sample (SRS) of size n from the underlying population, and measures the variable of interest for all sampled units. In the next step, for each of the n measured units of SRS, one draws an additional random sample of size $H - 1$ from the population to make n set samples of size H. Then, one ranks the H units in each set from smallest to largest. Finally, the judgment rank of the measured unit among its set is recorded. Therefore, the full data set consists of the n measured values X_i, and their judgment rank R_i, $i = 1, \ldots, n$, represented by independent and identically distributed pairs, $\{(X_i, R_i), i = 1, \ldots, n\}$.

The ranking process in this sampling scheme can be done by eye inspection, judgment of an expert, or using a concomitant variable. So, we will use the term judgment rank to emphasize that some errors may happen during the course of ranking and thus the ranking information may not be exact (imperfect ranking). If the ranking process is done without error, it is called perfect ranking, which is the ideal case.

A notable limitation of the empirical distribution function (EDF) is that, although it is commonly used to estimate the unknown cumulative distribution function (CDF) F, the resulting estimator is inherently discrete rather than smooth. Consequently, its derivative cannot be employed to make statistical inferences regarding functionals of the underlying probability density function (PDF). In addition, the EDF performs poorly in estimating the distribution function outside the range of the observed data, particularly in the tail regions beyond the sample extremes. These limitations motivate the development of smooth CDF estimators. Such estimators are important in a variety of applications, including reliability analysis, survival studies, environmental monitoring, and medical research, where accurate characterization of the underlying distribution is often required. Therefore, combining smooth CDF estimation with efficient sampling designs such as JPS provides a useful framework with both theoretical and practical significance.

This paper reviews smooth estimation methods for the CDF within the framework of JPS sampling developed by Azizi Kouhanestani et al. (2025). A general class of CDF estimators is presented, and their main finite-sample and asymptotic properties are summarized. The present conference paper aims to provide a concise and unified overview of these developments. In addition, an extended Monte Carlo simulation study is conducted to provide further empirical evidence on the finite-sample performance of the estimators. Unlike the original study, the present simulation includes bias evaluation, considers an additional probability distribution, and compares the EDFs under JPS and SRS designs, in addition to comparing the efficiency of the kernel-based CDF estimators.

2 A General Class of CDF Estimators in JPS

Let $X_1, \ldots, X_n$ be an SRS sample of size n from a population with an unknown CDF F. A general class of estimators of the CDF F is defined as

$$F_{n;srs}(t) = \frac{1}{n}\sum_{i=1}^{n} K_n(t - X_i),$$

where $K_n(.)$ is a given CDF. This estimator remains a valid CDF and, by selecting a continuous (or absolutely continuous) K_n, one obtains a smooth estimator that reflects the continuity properties of the underlying distribution function. Statistical properties of $F_{n;srs}(t)$ have been discussed by Yamato (1973).

As a particular case, setting $K_n(t) = e(t)$ leads to the classical unbiased EDF, $F^*_{n;srs}(t) = \frac{1}{n}\sum\limits_{i=1}^{n} \mathbb{I}(X_i \leq t)$, where $e(t) = \mathbb{I}(t \geq 0)$, and $\mathbb{I}(.)$ denotes the indicator function.

Consider a JPS sample $\{(X_i, R_i), i = 1, \ldots, n\}$ of size n from a population with an unknown CDF F. Let $\mathbf{R} = (R_1, \ldots, R_n)$ denote the ranks vector. Define indicator variables I_{ir} as follows: $I_{ir} = 1$ if X_i has judgment rank r ($R_i = r$), and $I_{ir} = 0$ otherwise, for $i = 1, \ldots, n$ and $r = 1, \ldots, H$. Thus, $N_r = \sum_{i=1}^{n} I_{ir}$ denotes the count of observations with judgment rank r.

Therefore, the variable $d_n = \sum_{r=1}^{H} \mathbb{I}(N_r > 0)$ counts the number of nonempty post strata formed during the JPS sampling procedure. Define $J_r = \frac{1}{N_r}$ if $N_r > 0$, and $J_r = 0$ otherwise, for $r = 1, \ldots, H$. Furthermore, note that the conditional distribution of each X_i given its judgment rank $R_i = r$, denoted by $F_{[r]}$, corresponds to the r-th judgment order statistic of a sample of size H ($X_{[r]}$).

Using the above notation, a general class of CDF estimators in the JPS setting proposed by Azizi Kouhanestani et al. (2025), is defined as

$$F_{n;jps}(t) = \sum_{r=1}^{H} W_r F_{n;[r]}(t), \tag{2.1}$$

where $W_r = \frac{\mathbb{I}(N_r > 0)}{d_n}$. Also, $F_{n;[r]}(t) = \frac{1}{N_r}\sum\limits_{i=1}^{n} K_n(t - X_i) I_{ir}$ if $N_r > 0$, and zero otherwise, where $K_n(.)$ is a given CDF.

If we take $K_n(t) = e(t)$, then the unbiased EDF in the JPS setting is obtained, and has the following form

$$F^*_{n;jps}(t) = \sum_{r=1}^{H} W_r F^*_{n;[r]}(t),$$

where $F^*_{n;[r]}(t) = \begin{cases} \frac{1}{N_r}\sum\limits_{i=1}^{n} \mathbb{I}\left(X_i \le t\right) I_{ir} & \text{if } N_r > 0, \\ 0 & \text{otherwise.} \end{cases}$

The properties of the EDF under the JPS sampling scheme were discussed by Dastbaravarde et al. (2016).

The following theorems, restated from Azizi Kouhanestani et al. (2025), summarize the main finite-sample and asymptotic properties of the proposed estimator given in equation (2.1). The corresponding proofs can be found in the original paper.

Theorem 2.1. *Let $\{(X_i, R_i), i = 1, \ldots, n\}$ be a JPS sample from a population with an unknown CDF F. Under a consistent ranking mechanism and a fixed set size H, every estimator of the form $F_{n;jps}(t)$ satisfies*

(a)

$$\mathbb{E}(F_{n;jps}(t)) = \frac{1}{H}\sum_{r=1}^{H} \mathbb{E}\big(K_n(t - X_{[r]})\big).$$

(b)

$$\begin{aligned}\mathbb{V}(F_{n;jps}(t)) &= H\mathbb{E}(W_1^2 J_1)\mathbb{V}\left(K_n(t-X)\right) \\ &\quad - \left[\mathbb{E}(W_1^2 J_1) - \frac{H}{H-1}\mathbb{V}(W_1)\right]\left(\sum_{r=1}^{H}\left(\mathbb{E}\left(K_n(t - X_{[r]})\right) - \mathbb{E}\left(K_n(t-X)\right)\right)^2\right).\end{aligned}$$

These results indicate that the bias of the JPS estimator coincides with that of its SRS counterpart. They also show that the JPS design yields an asymptotic variance that is at most equal to, and often smaller than, that obtained under SRS, even when ranking errors are present, provided the ranking process outperforms random assignment.

Theorem 2.2. *Let $\{(X_i, R_i), i = 1, \ldots, n\}$ be a JPS sample from a population with an unknown CDF F. Under a consistent ranking mechanism and a fixed set size H, every estimator of the form $F_{n;jps}(t)$ satisfies the following asymptotic results provided that $K_n \xrightarrow{w} e$ as $n \to \infty$, where $\xrightarrow{w}$ denotes weak convergence:*

(a)

$$\mathbb{E}(F_{n;jps}(t)) \to F(t).$$

(b)

$$\sup_{t\in\mathbb{R}} \left|F_{n;jps}(t) - F(t)\right| \xrightarrow{a.s.} 0,$$

where $\xrightarrow{a.s.}$ means almost sure convergence.

(c)

$$\sqrt{n}\left(F_{n;jps}(t) - F(t)\right) \xrightarrow{d} N\left(0, \frac{1}{H}\sum_{r=1}^{H} F_{[r]}(t)[1-F_{[r]}(t)]\right),$$

for all $t \in C(F)$ with $0 < F(t) < 1$, where $\xrightarrow{d}$ means convergence in distribution.

The above theorem shows that the considered class of estimators is asymptotically unbiased and strongly uniformly consistent. Furthermore, under the stated conditions, the estimators are asymptotically normal.

3 Kernel Distribution Function in JPS

This section considers the incorporation of smoothness constraints into the proposed JPS-based estimator in equation (2.1). To achieve this, the smoothing distribution function is assumed to depend on the sample size through a bandwidth parameter h_n, so that

$$K_n(t) = K\left(\frac{t}{h_n}\right),$$

where $K(\cdot)$ denotes a specified CDF and h_n is a smoothing parameter.

The function $K(\cdot)$ is generated from a symmetric kernel density $k(\cdot)$ with bounded support on $[-a, a]$. Specifically, the kernel satisfies

$$\int_{-a}^{a} k(x)\,dx = 1, \qquad \int_{-a}^{a} xk(x)\,dx = 0,$$

and has a finite nonzero second moment. Moreover, the bandwidth sequence is assumed to satisfy

$$h_n \to 0, \qquad nh_n \to \infty,$$

as $n \to \infty$. These conditions ensure that the bandwidth converges to zero but at a rate slower than $\frac{1}{n}$.

Under the above assumptions, the resulting estimator belongs to the class of kernel distribution function (KDF) estimators. For an SRS, the KDF estimator of F is given by

$$F^k_{n;srs}(t) = \frac{1}{n}\sum_{i=1}^{n} K\left(\frac{t - X_i}{h_n}\right).$$

For a JPS sample $\{(X_i, R_i),\, i = 1, \ldots, n\}$ drawn from a population with an unknown CDF F, the KDF estimator is given by

$$F^k_{n;jps}(t) = \sum_{r=1}^{H} W_r F^k_{n;[r]}(t),$$

where $F^k_{n;[r]}(t) = \frac{1}{N_r}\sum\limits_{i=1}^{n} K\left(\frac{t-X_i}{h_n}\right) I_{ir}$ if $N_r > 0$, and zero otherwise.

The efficiency of $F^k_{n;jps}(t)$ can be assessed through its mean squared error (MSE). It can be shown that

$$\begin{aligned} MSE\left(F^k_{n;jps}(t)\right) \approx H\mathbb{E}(W_1^2 J_1)\left(F(t)\left[1 - F(t)\right] - hf(t)\left(a - \int_{-a}^{a} K^2(u)du\right)\right) \\ + \frac{h^4}{4}\left(f'(t)\int_{-a}^{a} u^2 k(u)du\right)^2 - r(t), \end{aligned}$$

where $r(t)$ denotes the remaining terms which do not depend on the bandwidth h_n.

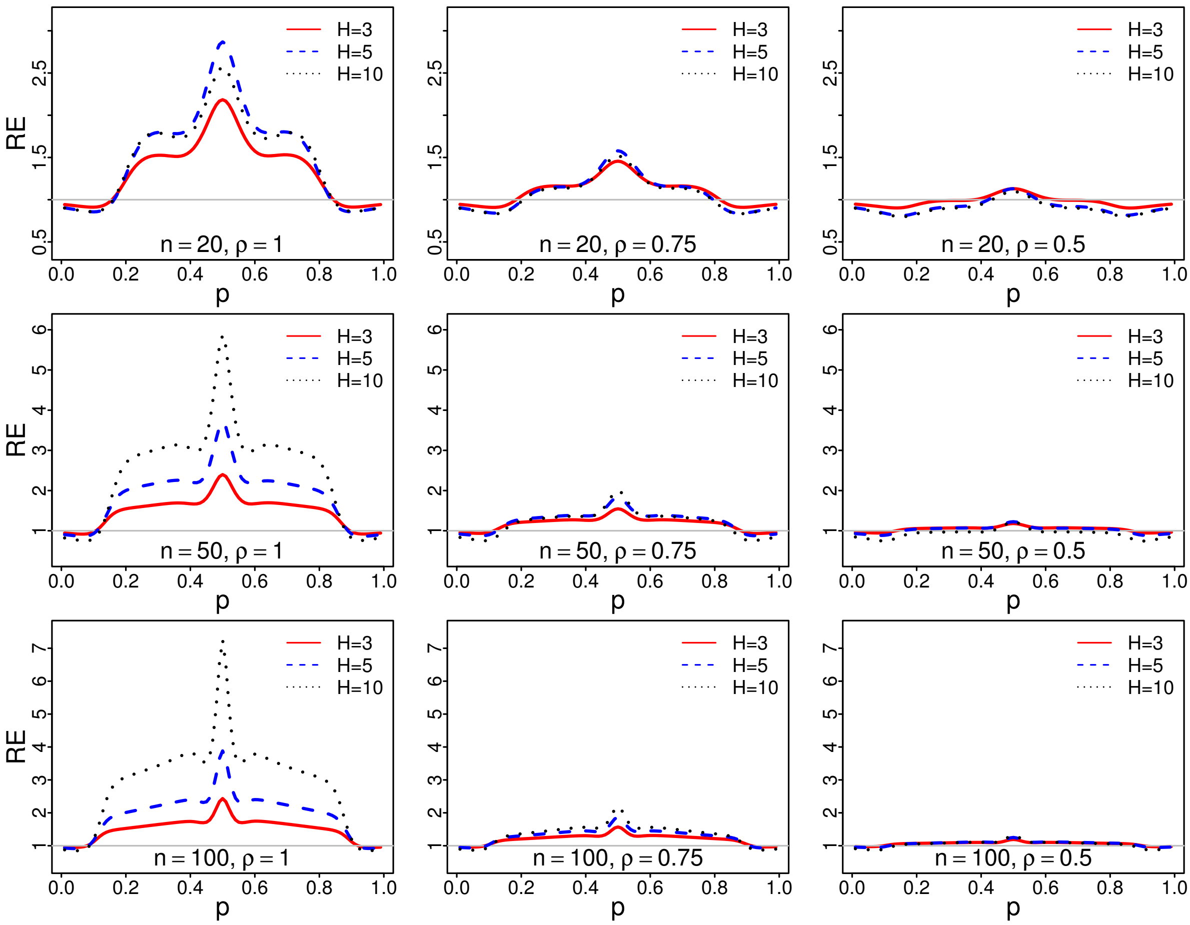

Figure 1: The simulated REs of $F^k_{n;srs}(.)$ to $F^k_{n;jps}(.)$ as a function of p, for $n \in \{20, 50, 100\}$, when $H = 3$ (represented by red and solid line), $H = 5$ (represented by blue and dashed line), $H = 10$ (represented by black and dotted line) and $\rho \in \{1, 0.75, 0.5\}$ under $U(-1, 1)$ distribution.

Therefore, the optimal bandwidth h_n in the JPS setting can be obtained by minimizing the MSE of $F^k_{n;jps}(t)$ with respect to h_n as

$$h_n = \left(\frac{nH\mathbb{E}(W_1^2 J_1) f(t) \left(a - \int_{-a}^{a} K^2(x)dx\right)}{n\left(f'(t)\int_{-a}^{a} x^2 k(x)dx\right)^2} \right)^{\frac{1}{3}}.$$

Since the resulting bandwidth expression involves the unknown quantities $f(t)$ and $f'(t)$, direct implementation is generally not feasible. A practical alternative is to employ the reference distribution approach proposed by Gulati (2004).

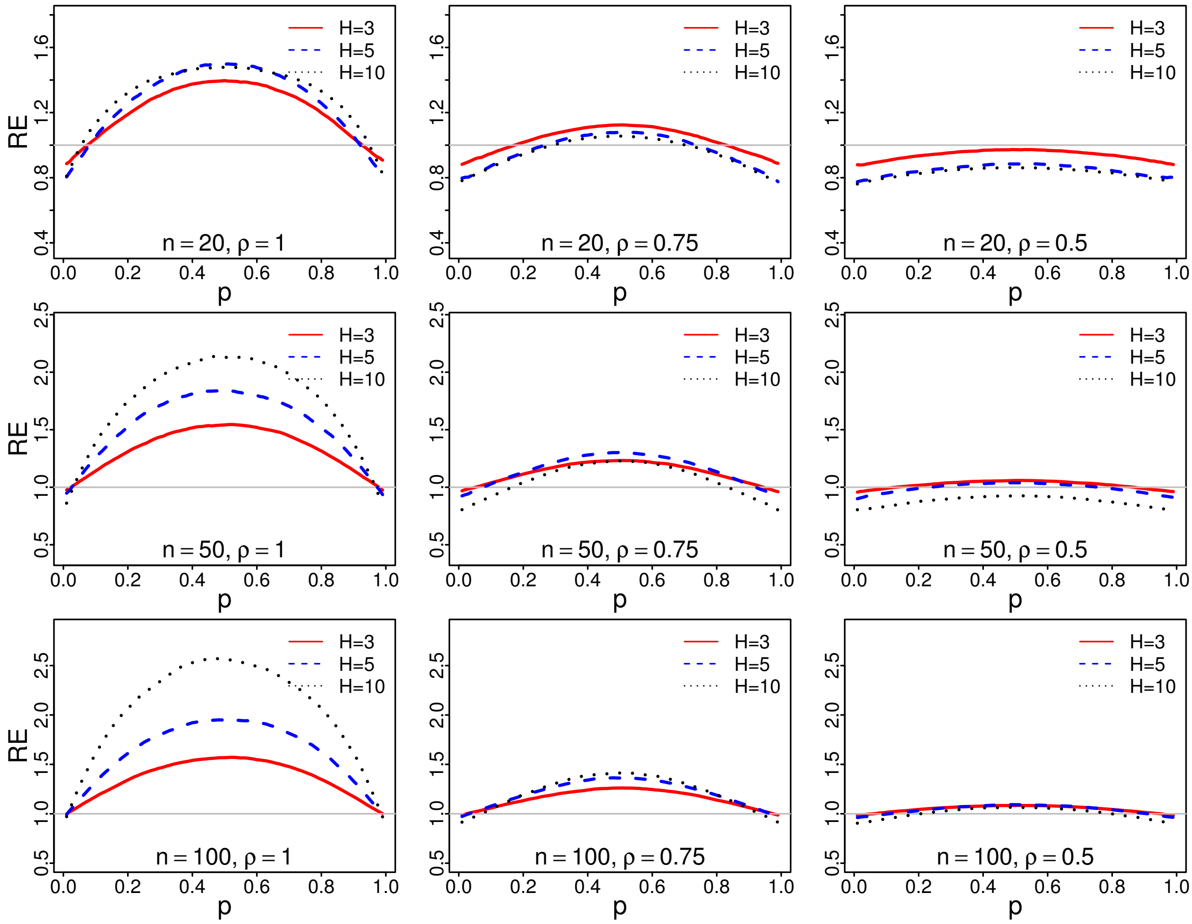

Figure 2: The simulated REs of $F^*_{n;srs}(.)$ to $F^*_{n;jps}(.)$ as a function of p, for $n \in \{20, 50, 100\}$, when $H = 3$ (represented by red and solid line), $H = 5$ (represented by blue and dashed line), $H = 10$ (represented by black and dotted line) and $\rho \in \{1, 0.75, 0.5\}$ under $U(-1, 1)$ distribution.

4 Monte Carlo Simulation

To evaluate the performance of the proposed KDF estimator, a Monte Carlo simulation study was carried out comparing the KDF estimator under the JPS design with its counterpart based on SRS. The comparison was conducted under both perfect and imperfect ranking scenarios. Sample sizes of $n \in \{20, 50, 100\}$ and set sizes of $H \in \{3, 5, 10\}$ were considered throughout the study.

Imperfect ranking was generated using the linear ranking error model of Dell and Clutter (1972). In this framework, ranking is performed through an auxiliary variable Y related to the study variable X by

$$Y = \rho \frac{X - \mu}{\sigma} + \sqrt{1 - \rho^2}\, Z,$$

where μ and σ^2 denote the mean and variance of X, respectively, and Z is an independent standard normal

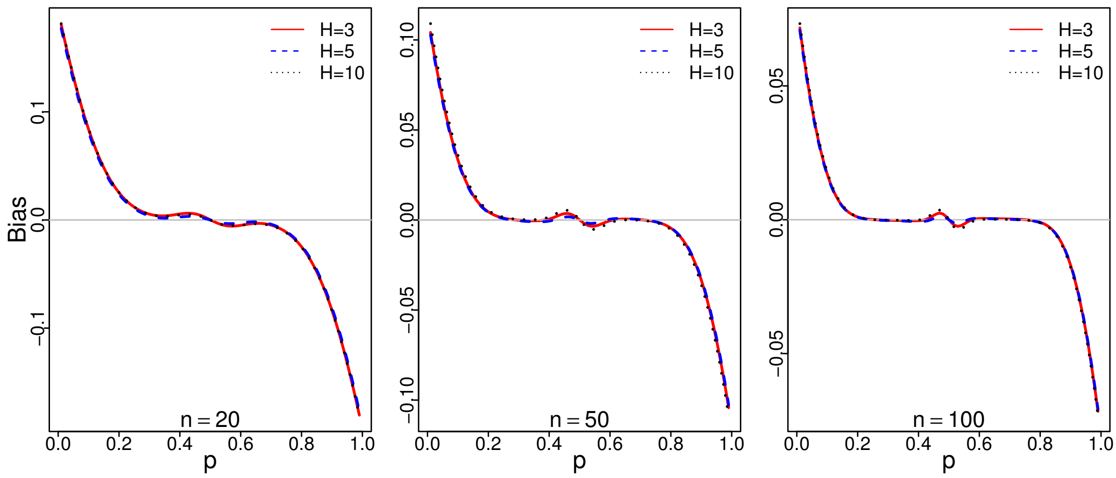

Figure 3: The simulated biases of $F^k_{n;jps}(.)$ as a function of p, for $n \in \{20, 50, 100\}$, when $H = 3$ (represented by red and solid line), $H = 5$ (represented by blue and dashed line), $H = 10$ (represented by black and dotted line) and $\rho = 1$ under $U(-1, 1)$ distribution.

random variable. The parameter ρ determines the ranking quality, with larger values corresponding to more accurate rankings. The values $\rho = 1$, $\rho = 0.75$, and $\rho = 0.5$ were selected to represent perfect, moderate, and weak ranking, respectively.

The efficiency comparison was based on the relative efficiency (RE) measure, defined as

$$RE(p) = \frac{\text{MSE}\big(F^k_{n;srs}(Q_p)\big)}{\text{MSE}\big(F^k_{n;jps}(Q_p)\big)},$$

where Q_p denotes the p-th quantile of the underlying distribution. Values of $RE(p)$ greater than one indicate that the JPS-based estimator outperforms its SRS analogue at the corresponding quantile.

For each combination of (n, H, ρ), 100,000 samples were generated from the uniform distribution on $(-1, 1)$ under both sampling schemes, and the corresponding REs were computed for $p = 0.01, 0.02, \ldots, 0.99$. The KDF results based on the truncated Gaussian kernel are presented in Figure 1.

From Figure 1, we find out that the REs are an increasing function of sample size n. The efficiency of the JPS estimators is usually higher around the center of the parent distribution and lower at its boundaries. Furthermore, the REs increase with the ranking quality ρ.

Also, we observe that the optimal value of the set size H, which leads to a higher efficiency, depends on both ranking quality (ρ) and sample size (n). Specifically, a larger value of set size H leads to a higher efficiency provided that the sample size is large enough and the ranking quality is sufficiently good. Therefore, small values for set sizes ($H = 3, 5$) are recommended to be used in practice if the sample size is small or there is any doubt about ranking quality. The corresponding EDF results are also shown in Figure 2, and

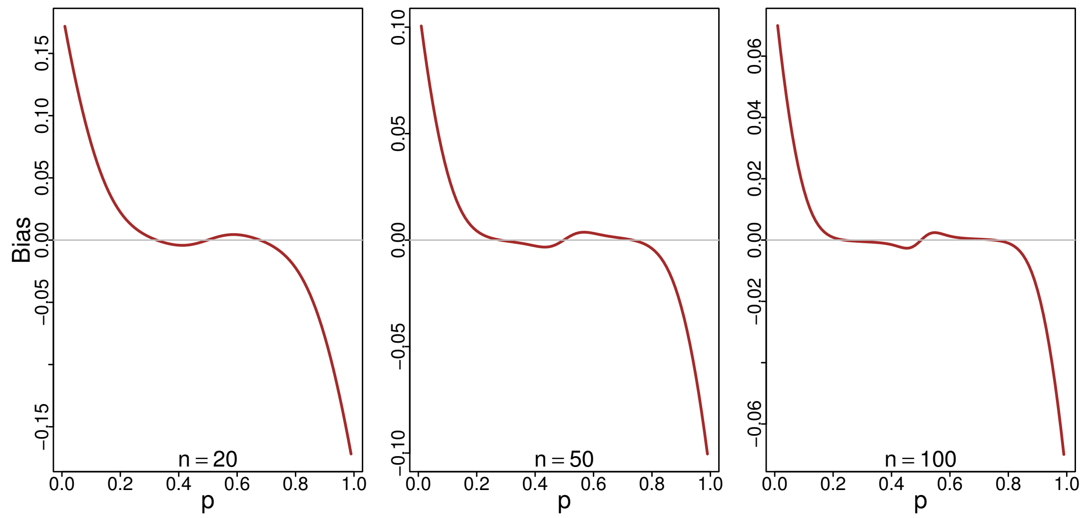

Figure 4: The simulated biases of $F^k_{n;srs}(.)$ as a function of p, for $n \in \{20, 50, 100\}$, under $U(-1, 1)$ distribution.

their interpretation is similar to that of the KDF results.

It is well known that the EDF is an unbiased estimator of the CDF under both SRS and JPS sampling schemes. Therefore, in addition to comparing the MSEs, we investigated the finite-sample bias of the kernel-based CDF estimator under both SRS and JPS in the perfect ranking scenario. Since the theoretical bias of the estimator does not depend on the ranking quality, this investigation was conducted under the perfect ranking scenario. The results are shown in Figures 3 and 4. As expected from the theoretical results, similar patterns are observed under both the SRS and JPS sampling schemes.

We also observe that the magnitude of the bias decreases as the sample size increases across all considered scenarios. This behavior is consistent with theoretical expectations, since the bias of the kernel-based estimator diminishes as the bandwidth decreases with increasing sample size.

The results indicate that the improvement in MSE achieved under JPS is not attributable to bias reduction. Rather, the gain arises primarily from variance reduction resulting from the additional ranking information incorporated into the JPS design. This advantage persists even when the ranking quality is imperfect, as JPS continues to provide a modest reduction in variability relative to SRS.

Overall, these findings suggest that the practical advantage of JPS over SRS arises from utilizing the additional information provided by the ranked sample, which improves the precision of smooth CDF estimation for a given sample size.

Discussion and Conclusion

In this paper, we reviewed the general class of estimators for the population CDF in the JPS framework, encompassing both the EDF and KDF. We compared the performance of the EDF and KDF estimators under JPS sampling with their counterparts under SRS using Monte Carlo simulation. It was found that the JPS estimators outperforms its SRS competitor in most of the considered cases.

Future research in this area may focus on extending smooth CDF estimation under JPS sampling to multivariate settings and developing adaptive bandwidth selection approaches. Such extensions could further improve the flexibility and applicability of smooth CDF estimation methods under ranked sampling designs.

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